

Cover

QEM-9: Hyperstacked Mirror Lattices under the Opposition Principle (v2.1)

QEM-9超堆叠镜像晶格下的反对原则v2.1

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Abstract (EN)

Abstract. We extend the Quantum Entanglement Mechanics (QEM) framework from 3D mirror lattices to d-dimensional (and asymptotically infinite-dimensional) hyper-stacked lattices by enforcing a Σ/Δ Opposition Principle. Each stacking axis is assigned a Z_2 opposition operator O_k ; the global opposition $O^{(d)} = \otimes_k O_k$ confines admissible modes to the odd (Δ) sector. Within this sector we derive: (i) a sector-resolved upper bound on propagation speed, $v_{\max}^{(d)} \approx \min(v_{\text{local}}, (1/\pi)\sqrt{(a_{\parallel}^2 \Omega^2 + \sum_k a_k^2 c_k^2 \lambda_{\max,k}^{\text{odd}})})$; (ii) stability islands with delay limits $\tau_c(d) < 0.2 \cdot \min_k \{ \xi_k / c_k \}$; and (iii) a coherence scaling $\xi_d \sim \xi_1/\sqrt{d}$. We prove controllability as $d \rightarrow \infty$ provided $\sum_k c_k^2$ is bounded. Engineering rules are delivered as a design-for-opposition SOP: bipartite layouts, loop-parity (even Σ count), π -inverters on odd- Σ paths, and sector-selective validation (parity-echo, basis-flip, sector spectroscopy). The results unify QEM-4...7 and yield practical DRC-style constraints for scalable quantum/photonic/atomic architectures.

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Novelty and Related Work

Novelty and Relation to Prior Work.

- (1) We formalize a Σ/Δ Opposition Principle across stacking axes (Z_2 operators O_k) and restrict transport/stabilization to the odd (Δ) sector.
- (2) We derive d-dimensional propagation bounds and stability islands with delay limits in the odd sector, yielding closed-form scalings: $v_{\max}$ with a sum-of-squares under a square-root, $\tau_c(d) \sim 1/\sqrt{d}$, and $\xi_d \sim \xi_1/\sqrt{d}$.
- (3) We provide a design-for-opposition SOP/DRC for real chips. Prior works study components individually (graph-product eigenvalue additivity; bipartite no-odd-cycle; π phase for negative couplings; Lieb–Robinson-type bounds). Here we synthesize them into a unified, sector-resolved, engineering-ready framework for hyper-stacked entanglement.

Appendix: Engineering SOP

Design-for-Opposition SOP (DRC-style).

- (1) Bipartite layout (two-color nodes; edges connect only opposite colors).
- (2) Loop parity: every closed loop carries an even number of Σ links (opposition closure).
- (3) π -inverters on odd- Σ paths to restore Δ -parity at endpoints.
- (4) Delay budget per axis: $\tau_k < 0.2 \cdot \xi_k / c_k$; prefer localized feedback; fan-out limited by $||W||$.
- (5) Scalability: ensure $\sum_k c_k^2$ is bounded (e.g., $c_k \sim 1/\sqrt{d}$ or $1/d$).
- (6) Validation: parity-echo, basis-flip, sector spectroscopy targeting the Δ sector.